Replication of Kopf (2013, PLoS One)

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Project links

- **Original paper:** Kopf, J. et al. (2013). The Effect of Emotional Content on Brain Activation and the Late Positive Potential in a Word n-back Task. *PloS One*. 2013 Sep 26;8(9):e75598. https://doi.org/10.1371/journal.pone.0075598
- **Preregistration:** OSF registration *(registered YYYY-MM-DD, before final data collection)*
- **Live experiment:** < https://psych251.github.io/kopf2013/> (take the study exactly as participants did)
- **Repository:** https://github.com/psych251/kopf2013 (experiment code, data, and this report)

AI use statement

Tool and model	What it was used for	How you checked the output
ChatGPT(GPT-6.1 Sol Medium)	Brainstormed project ideas and relevant papers to choose to replicate given my personal research interests	Verified citations produced by ChatGPT by manually searching on Google Scholar; compared suggested papers with personal selections; read through each paper
ChatGPT(GPT-6.1 Sol Medium)	Helped edit code in R for creating key statistics and power analysis table	Confirmed output via manual code review
ChatGPT(GPT-6.1 Sol Medium)	Searched for English language databases to use rated on valence and arousal	Downloaded relevant papers and chose paper to use English word list from
ChatGPT(GPT-6.1 Sol Medium)	Generated list of 30 English words to use as stimuli for study, matching words from Warriner et al. to Berlin Affective Word List from original study (z-scored both scales, then matched as similar as possible)	Cross-checked valence numbers with CSV file downloaded as supplementary file from Warriner et al.; manually checked word/syllable length
ChatGPT(GPT-6.1 Sol Medium)	Generated list of 30 neutral English words to use as stimuli for the practice study, using words fro Warriner et al.	Cross-checked valence numbers with CSV file downloaded as supplementary file from Warriner et al.; manually checked word/syllable length

Tool and model	What it was used for	How you checked the output
Claude(Opus 5.5 Medium)	Edited experimental paradigm from demo to fit paradigm from original study as closely as possible	Manually reviewed code before committing to github in a separate branch, then tested the study as a participant before merging branch to main
Claude(Opus 5.5 Medium)	Added temporary ‘debug’ buttons to skip thorugh experiment for debugging purposes; will be inactivated before actually deploying study	Manually reviewed code and went through github pages link to make sure the buttons functioned as expected.
ChatGPT(GPT-6.1 Sol Medium)	Edited analysis code sections of replication report to match analysis structure of my study	Manually reviewed code and ensured output was as expected
ChatGPT(GPT-6.1 Sol Medium)	Added ‘data report’ section to write-up to summarize results of pilot b	Manually reviewed code and ensured output was as expected

I wrote all of the text in this report myself. AI tools were not used to write, rewrite or paraphrase it, and were not used to generate, alter, or impute any participant data. I take full responsibility for the content of this report, including any errors introduced by AI tools.

Key statistics

▶ Show the analysis code (shared by Phase 1 and Phase 2)

Key comparison: valence × difficulty interaction on error counts in a 3 × 3 repeated-measures ANOVA. Original degrees of freedom are Greenhouse–Geisser corrected. Replication results use the correction when Mauchly’s test indicates violated sphericity ($p < .05$).

Study	N	ANOVA comparison	F	df1	df2	p
Original	30	valence * difficulty	3.70	2.80	82.10	0.016
Phase 1: replication	5	valence * difficulty	0.43	4.00	16.00	0.78
Phase 2: extension		valence * difficulty				

Phase 1: Replication

Introduction

The empirical article by Kopf et al. “The effect of emotional content on brain activation and the late positive potential in a word n-back task,” published in PLoS One in 2013, tests the hypothesis that emotional words, particularly the valence of those words, impact working memory performance. The authors found that across different blocks of the n-back task where the stimuli consisted of either positive, negative, or neutral words (which varied only by valence level, not by arousal level), more errors were made on difficult tasks when the valence was negative. I plan to replicate the main finding of the paper displayed in Figure 2 regarding errors made in the n-back test in different valence categories. This finding has relevance for my broader research interests that span not only the fundamental mechanisms underlying cognitive control, but also the factors that influence cognitive control performance, including mental health conditions and general emotional states. My first-year project involves investigating subcortical brain regions involved in a variety of cognitive control domains; one of the tasks in the dataset I am utilizing from my lab is a version of the n-back, the same task utilized in this study. Replicating this project will allow me to gain more foundational knowledge of this task paradigm as I implement the design in Javascript and motivate me to consider key considerations of how to analyze behavioral results from this task.

The procedure described in the paper involves using words from the Berlin Affective Word list, which are classified for arousal and valence, and also matched for properties like word length, as stimuli in various blocks of the n-back task. There were 27 blocks in the task, three blocks of each of the conditions described next: a 1-back task with negative-valenced words as stimuli, a 2-back task with negative-valenced words as stimuli, and a 3-back task with negative-valenced words as stimuli; a 1-back task with neutral-valenced words as stimuli, a 2-back task with neutral-valenced words as stimuli, and a 3-back task with neutral-valenced words as stimuli; a 1-back task with positive-valenced words as stimuli, a 2-back task with positive-valenced words as stimuli, and a 3-back task with positive-valenced words as stimuli. A challenge I forsee is matching the words from the Berlin Affective Word list with English-equivalent words. I will have to find a similar word list that rates words on valence and arousal, as the English-translated versions of the words used in this study might not be sufficient, or might not be matched on word length, etc.

The key statistical test I will pre-register is collect each participant’s error rates on the blocks, then run a 3 (emotional valence of word) * 3 (difficulty level of task) repeated ANOVA. The test is the interaction effect of word valence on error rate. The original study reported the results as such: “A significant interaction effect for the errors ($F(2.8, 82.1) = 3.7$, $p = .016$) was observed, indicating that word valence influenced the error rate differentially in each difficulty level, the more difficult the task; the more errors were made especially in the negative valence condition.” A possible extension I might complete for this study if it does not replicate is trying to recruit German speakers on Prolific to complete the task, and use the exact same words from the Berlin Affective Word list as in the original paper. I will use a first-language screener in Prolific (English), and screen for U.S. residents, which might take a bit longer to recruit people to complete the task.

Methods

Power analysis

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Approximate sample needed for the valence × difficulty interaction, assuming the original standardized effect (partial $f^2 = 0.126$) and Greenhouse–Geisser epsilon = 0.71, alpha = .05. All participants complete all nine conditions.

Power	N per condition	N total
80%	33	33
90%	42	42
95%	50	50

The original sample size was intended to be 32 participants, but 2 were excluded, for a final sample size of 30 participants. The original effect size was F(2.8, 82.1) = 3.7 (which we can calculate as partial f2 = 0.126). The table above shows the sample size needed to detect this effect size. In this instance, 2.5 times the original sample would be 75 participants (per condition and total, since this is a within-participant study). We will use the 2.5 times original sample method and recruit 75 participants. Since the task takes about 25 minutes and is $10 per hour, the budget will be around $312.50 for the first replication.

Planned sample

The planned sample is US residents on Prolific, fluent in English, 18 or older. There are no filtering questions for the first replication. The planned sample size is 75 participants, following the 2.5* original sample rule that is a widely-used alternative to using the original sample size (Simonsohn, 2015). I will collect until 75 participants pass the exclusion criteria described below in the Analysis plan (i.e., 75 participants that actually completed the experiment correctly and paid attention).

Materials

- The n-back task structure is identical to the original article, quoted here: “Participants saw blocks of ten consecutive words presented on a computer screen. Two of these words were used as cues and were followed by a target word which was presented one, two or three words after the cue word, to vary the difficulty level and thus the working memory load. When the participants saw the target word, they were instructed to press the space key on a standard keyboard. Each block of ten words comprised one difficulty level. Also the emotional content of the words was varied; the words were positive, negative or neutral. Each block encompassed only one emotional valence. In total, there were three blocks where the emotional valence was negative and the difficulty level was one, meaning the target word followed the cue word directly. Three blocks had a negative emotional valence and the difficulty level was two, meaning there was one word between cue word and target word. Three blocks had a negative emotional valence and the difficulty level was three, meaning there were two words between the cue word and the target word. The same was true for the positive and the neutral emotional valence; therefore, in sum the experiment had 27 blocks. Each block was 20 seconds long, and blocks were separated by a 20-second pause. Within each block, each word was visible for 500 ms, followed by a blank screen, which was also 500 ms long. The difficulty level of the blocks was pseudorandomized, and difficulty levels always varied from one block to another. In the pause preceding a block, the instruction for the task (1-, 2-, 3-back) was presented on the screen.”

- The words in the original study were as follows: “The words (a list of which can be seen in Table 1) were chosen from the Berlin Affective Word List [11], were all nouns and matched for valence (positive nouns had a mean valence of 2, negative nouns a mean valence of -2 and neutral words a mean valence rating of 0, with the valence scale spanning from -3 for very negative to 3 for very positive). Mean arousal of emotional words was matched in respect of their valence, meaning the negative and positive words had a mean arousal of 3, the neutral words had a mean arousal of 0 (on a scale from 1: very low arousal to 5: very high arousal). To

ensure equal working memory load for all words, phoneme length (5), syllable length (2), and number of letters (6) were matched as well." Negative words with their English translations were: "TYPHUS (typhus), BEFEHL (order), ARREST (warrant), TYRANN (tyrant), GREUEL (horror), SEENOT (distress at sea), HORROR (horror), TRAUMA (trauma), TERROR (terror), and UNHEIL (calamity)". Positive words were "RETTER (savior), WISSEN (knowledge), MUTTER (mother), GEWINN (win), GEFÜHL (feeling), HIMMEL (heaven), FRIEDE (peace), URLAUB (vacation), SOMMER (summer), and FREUND (friend)". Neutral workds were: "ABLAUF (sequence), AFFEKT (affect), BANNER (banner), GERUCH (smell), KELLER (basement), LOSUNG (watchword), STELLE (position), ZEUGIN (eye witness), AKZENT (accent), and INHALT (content)."

- We used as close to equivalent words in English as possible, matched on the same valence/arousal conditions and matched on phoneme/word length as in the original study. Words were taken from Warriner et al.'s "Norms of valence, arousal, and dominance for 13,915 English lemmas" (https://doi.org/10.3758/s13428-012-0314-x) (downloaded supplementary CSV file with words and their ratings). We asked Chatgpt (GPT-6.2 Sol Medium) to come up with a list of words with similar arousal/valence as the German words in the original study (z-scored the scales to then compare their ratings), and matched on word length, etc., then verified the output by checking the CSV itself and the original paper to see if it matched the original study. Please see the tables below for the word lists. Note that English words use a 1-9 scale, as opposed to the -3 to +3 valence scores of the Berlin Affective Word List, which is why we had ChatGPT do z-scoring for the comparison with the German word list.

Negative word	Valence	Arousal	Valence z	Arousal z
trauma	2.89	4.87	-1.705	0.736
deceit	2.89	5.62	-1.705	1.572
lesion	2.95	5.62	-1.658	1.572
maggot	2.00	5.60	-2.403	1.550
hazard	2.90	5.77	-1.697	1.740
danger	2.66	5.84	-1.886	1.818
famine	2.95	6.10	-1.658	2.108
battle	3.52	6.25	-1.211	2.275
spider	3.35	6.91	-1.344	3.011
zombie	3.57	6.53	-1.172	2.587

Positive word	Valence	Arousal	Valence z	Arousal z
access	6.68	5.30	1.268	1.215
orchid	7.00	4.11	1.519	-0.112

Positive word	Valence	Arousal	Valence z	Arousal z
muffin	7.10	4.05	1.597	-0.179
riches	7.10	5.42	1.597	1.349
yogurt	7.00	4.52	1.519	0.345
season	6.71	3.20	1.291	-1.127
vision	6.79	3.18	1.354	-1.150
silver	7.48	4.00	1.895	-0.235
cuddle	7.60	3.83	1.989	-0.425
kitten	7.58	3.19	1.974	-1.139

Neutral word	Valence	Arousal	Valence z	Arousal z
domain	5.05	3.36	-0.011	-0.949
cannon	4.74	5.09	-0.254	0.981
module	4.90	4.33	-0.129	0.133
fiddle	5.05	4.05	-0.011	-0.179
gender	5.05	4.05	-0.011	-0.179
method	5.20	4.17	0.107	-0.045
duffel	4.90	3.50	-0.129	-0.793
tassel	4.95	4.78	-0.089	0.635
copter	4.86	4.70	-0.160	0.546
retail	5.00	3.62	-0.050	-0.659

Category	Mean valence	Mean arousal	Mean valence z	Mean arousal z
Negative	2.968	5.911	-1.644	1.897
Positive	7.104	4.080	1.600	-0.146
Neutral	4.970	4.165	-0.074	-0.051

Procedure

- First, the participant views the consent screen and consents to participate in the study.
- Second, the participant completes a demographics page.
- Third, the participant sees the welcome page orienting them to what they should expect from the study.
- Fourth, the participant goes through a series of instruction screens where they are instructed how to do a 1-back, 2-back, and 3-back task. They are then presented with a question that checks their understanding of the instructions. A correct answer will lead to a 'correct' screen, whereas a wrong answer will send them back to instructions.
- Fifth, the participant enters the practice block, which follows the same structure as the original study: "Before the experiment, participants practised the task using one block of each difficulty (1-, 2-, 3-back). Before the task itself, completed a training phase, where they saw 30 consecutive neutral words in the 1-back condition, the same neutral words in the 2-back condition and in the 3-back condition."
- Sixth, the participant enters the main task block, which follows the same structure as the original study: "Participants saw blocks of ten consecutive words presented on a computer screen. Two of these words were used as cues and were followed by a target word which was presented one, two or three words after the cue word, to vary the difficulty level and thus the working memory load. When the participants saw the target word, they were instructed to press the space key on a standard keyboard. Each block of ten words comprised one difficulty level. Also the emotional content of the words was varied; the words were positive, negative or neutral. Each block encompassed only one emotional valence. In total, there were three blocks where the emotional valence was negative and the difficulty level was one, meaning the target word followed the cue word directly. Three blocks had a negative emotional valence and the difficulty level was two, meaning there was one word between cue word and target word. Three blocks had a negative emotional valence and the difficulty level was three, meaning there were two words between the cue word and the target word. The same was true for the positive and the neutral emotional valence; therefore, in sum the experiment had 27 blocks. Each block was 20 seconds long, and blocks were separated by a 20-second pause. Within each block, each word was visible for 500 ms, followed by a blank screen, which was also 500 ms long. The difficulty level of the blocks was pseudorandomized, and difficulty levels always varied from one block to another. In the pause preceding a block, the instruction for the task (1-, 2-, 3-back) was presented on the screen."
- Seventh, the participant sees a screen with an optional free-text box where they can write any feedback about their participation in the study.
- Last, the participant sees the debrief screen with a 'Finish' button that saves the data. The entire task is expected to take around 25 minutes.
- The only exception between our study and the original study is adding in particular consent, demographics, feedback, and debrief screens. The task was the same except the words were in English, as described above.

Analysis plan

Key analysis of interest. On more difficult n-back tasks, more errors are made on blocks with negatively-valenced words as stimuli. This is tested as a 3 by 3 (emotional valence of word (positive, negative, neutral) by difficulty of task (easy, medium, hard/1-back, 2-back, 3-back)) repeated within-subjects ANOVA. The reported statistics include the F-value with degrees of freedom and the p-value on the interaction effect of word valence on error rate (the degrees of freedom are corrected with Greenhouse-Geisser method if sphericity is violated as described in the original paper).

Exploratory analysis. We also looked at the n-back response times based on valence, as described in the Exploratory analyses section below.

Exclusions. The original paper does not report exclusion reasons, other than 2 of the participants were excluded "for technical reasons." I will add additional behavioral checks to exclude participants who appear to have not paid attention at all during the study or attempt to cheat. I will exclude on the conditons as follows: Participants who never press the response key during any block of the n-back task; participants who respond on more than 80% of non-target trials (i.e., the word they see is NOT the same as the word n trials before)

Differences from the original study

Aspect	Original study	This replication	Expected to matter?
Sample	Thirty-two healthy right-handed volunteers (in person), German speakers	Adults recruited on Prolific, English speakers from the U.S.	Possibly. Even if our English words matched the Berlin Affective Word list exactly, cultural differences between German and U.S. participants could elicity varying results
Setting	Online task but conducted in person; participants wore fNIRS equipment and EEG electrodes while completing task	Online, unsupervised, on the participant's own computer, without fNIRS or EEG equipment	Possibly. Wearing the equipment (which is likely uncomfortable) could have some influence on participants' overall n-back task performance
Wording	German words from Berlin Affective Word list	English words from Warriner et al.	Unlikely. English words matched conditions of words from original study; they were matched on word and syllable length, arousal, and words were chosen to have equivalent valences to the German word set used in the original study.
Design	Within subjects, pseudo-randomized condition order	Within subjects, pseudo-randomized condition order	The study specifies a psuedo-randomized condition order whereby the difficulty level had to vary from one block after the other, so in the replicated study the exact order for each subject will

Aspect	Original study	This replication	Expected to matter?
			be different than the exact order for each subject in the original study. This is unlikely to have any effect.
Response	Keyboard press (space key on standard keyboard)	Keyboard press (space key; however we cannot regulate what kind of keyboard participants have access to at home)	Unlikely. Most participants will probably have a standard keyboard, and any keyboard has a space bar on it.
Exclusion criteria	Not specified other than “technical reasons”	Excluded based on participants who never pressed response key during any n-back block, or responded on more than 80% of non-target trials	Unlikely. If the original study included participants who weren’t paying attention, then it would have likely made the effect size less significant. However, the original study took place in person, so experimenters would have likely noticed if a participant wasn’t paying attention (these checks matter more for our online study on Prolific).
Other tasks	Tasks were completed alongside fluid and crystallized intelligence questionnaires, a clinician interview to assess for neuropsychiatric diagnoses, and a emotions/mood questionnaire (PANAS).	Instructions were followed by an instruction comprehension check, all additional surveys were excluded	Unlikely: the instruction check is meant to ensure participants understand the task, not to change any results. Given the within-subjects design for the behavioral result, the additional surveys (intelligence, mood, neuropsychiatric diagnoses), which were included in original study to confirm that there were comparable intelligence

Aspect	Original study	This replication	Expected to matter?
			levels in the sample and that there were no neuropsychiatric diagnoses, should not have any effect on within-subjects ANOVA results.

Methods addendum (post data collection)

Guidance

Leave this section out of the version you preregister. After data collection, report the actual sample and any deviation from the plan above.

Actual sample

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Participants at each step. All analysed participants complete the within-subjects valence × difficulty design.

Step	Total
Started	9
Completed	6
Passed exclusion criteria	5

Of 9 participants who started, 6 completed the study and 5 were retained after exclusions. 1 completed session(s) contained no response during the main n-back task and were excluded.

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Age of the analysed sample.

N	Mean age	SD age	Age not given
5	33.8	16.6	0

Data report

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Time through each participant's last saved trial, including instructions, practice, and breaks.

Participant ID	Completed	Recorded time (minutes)

Participant ID	Completed	Recorded time (minutes)
E4xlD324VnXrcBt1ITbSrMkUqqM2-2bb3a418	Yes	23.64
IWXO6LtqBMViyY3ORHnjnSXYDp22-dd6a4545	Yes	29.09
KfhihOmpPbPVw8B6fAgtLXL4aBE3-68faa9a3	Yes	24.11
XVqW30o9fMYknZQTcZt4o8kgZsy1-f52c1a4a	Yes	24.25
aDWKhqcjrGcbPcheBbuMbd7r6qa2-8a83eaa7	No	0.86
lW3CnljfypWutclxendUNtZx8oi2-6f274524	No	3.88
mAMCjpuu7AhgHrHxoblW3jO4APx2-50a7bd87	Yes	21.55
quwi1drHHOOclHKO0VxD2pASHDf1-98c8e643	No	6.51
sp3qVWaUS4e26S3RwdQ7iOnRFr12-0b729690	Yes	65.66

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Average recorded completion time.

Completed participants	Participants with timing data	Mean time (minutes)	SD (minutes)
6	6	31.38	16.98

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Non-completion at export; this may include sessions still in progress.

Started	Completed	Not completed	Dropout rate
9	6	3	33.3%

After pilot B, the average time elapsed was 31 minutes. However, most people who completed the study took 21-29 minutes, most under 25, and one person just took 65 minutes. This was an outlier, so I will keep the timing the same. It seemed that people comprehended the experiment. In terms of feedback, one person indicated the n-back task was difficult to follow for higher n, and someone else said to give a shorter wait time in between blocks, but to remain faithful to the original study, no further changes will be made.

Deviations from the preregistered plan

Guidance

Any deviation from the preregistered plan above, or “None”. Changes made after seeing data belong here even if they seem minor, together with the reason.

Results

Data preparation

Confirmatory analysis

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Confirmatory analysis: valence × difficulty interaction on error counts in a 3 × 3 repeated-measures ANOVA. Greenhouse–Geisser correction is applied when Mauchly's test indicates violated sphericity ($p < .05$).

N	ANOVA comparison	F	df1	df2	p
5	valence × difficulty	0.43	4.00	16.00	0.78

Replication assessment

Guidance

Judge the replication on effect sizes, not only on whether the key test is significant. Compare the replication estimate and its interval with the original: does the interval exclude zero, and does it include the original estimate? A significant result much smaller than the original, or a non-significant result whose interval still includes the original effect, are both informative and should be reported as such (Experimentology §3.2).

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Valence × difficulty interaction on error counts. Original degrees of freedom are Greenhouse–Geisser corrected. Replication results use the correction when Mauchly's test indicates violated sphericity ($p < .05$). When Mauchly's test cannot be estimated, uncorrected replication results are shown for pilot debugging only. Original partial eta squared is reconstructed from rounded published statistics.

Study	N	Partial eta squared	F	df1	df2	p
Original	30	0.112	3.70	2.80	82.10	0.016
Phase 1: replication	5	0.098	0.43	4.00	16.00	0.783

The replication's valence × difficulty interaction had partial eta squared = 0.098, compared with approximately 0.112 in the original study. The replication interaction was not statistically significant at alpha = .05 ($p = 0.78$).

▶ Show code

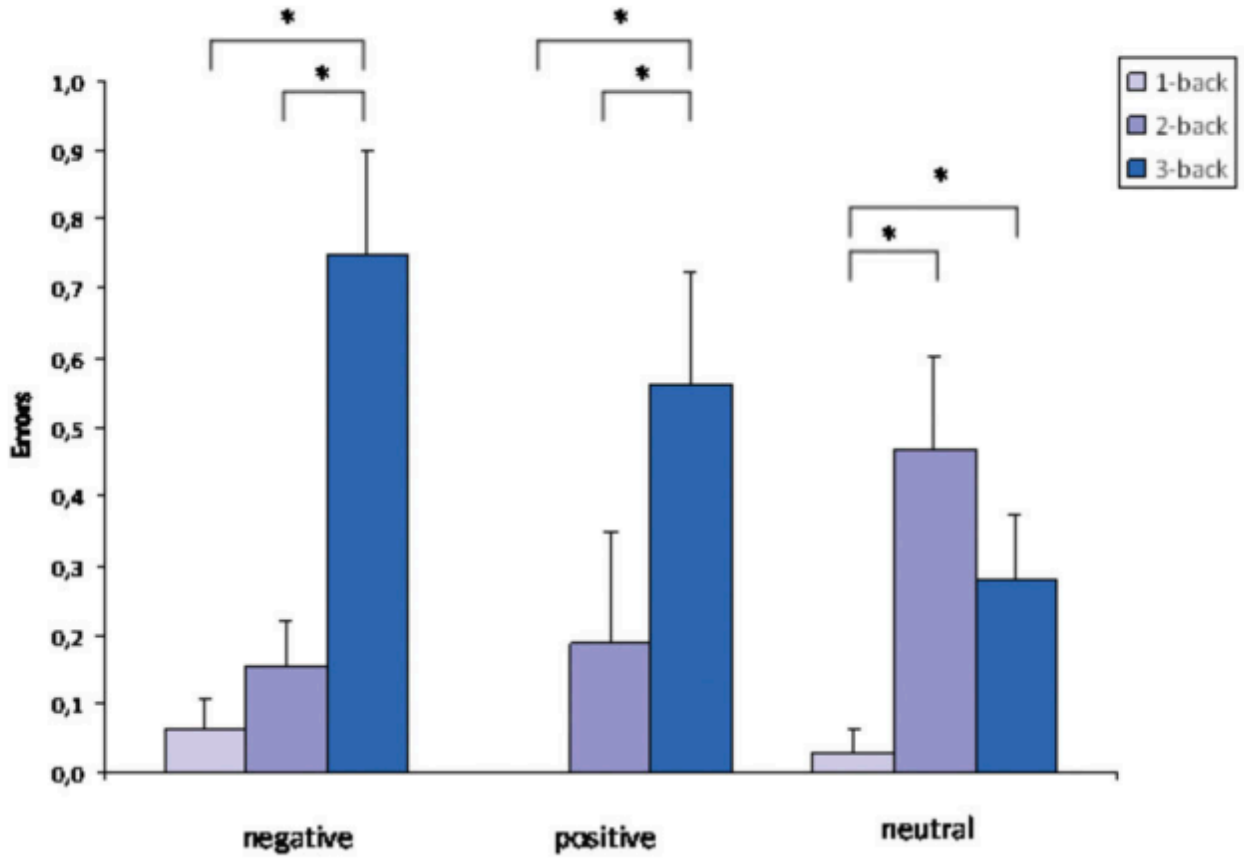


Figure 2. Number of errors. Errors made in the emotional n-back test in the different valence categories. Error bars indicate the standard error. An asterisk indicates a significant result.

Original: Kopf et al. (2013), Figure 2. Error bars indicate standard errors. Source: doi:10.1371/journal.pone.0075598.g002.

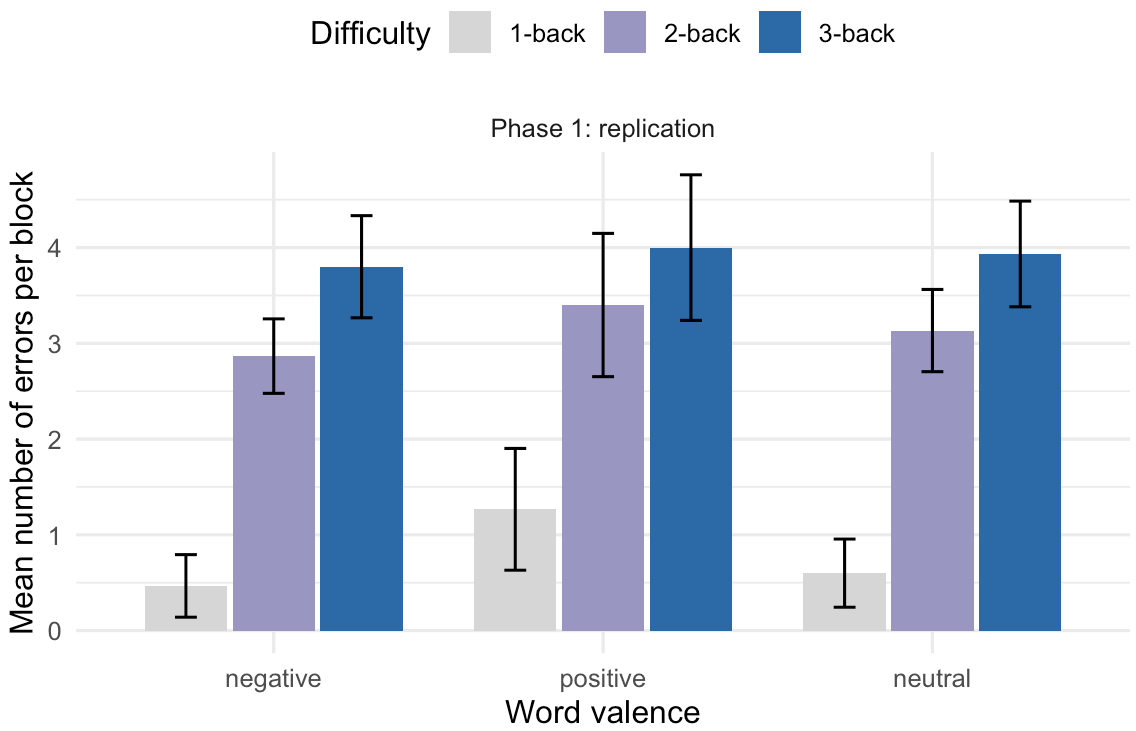


Figure 1: Phase 1 replication: mean errors per block by word valence and task difficulty. Error bars indicate ±1 SE across participants.

Exploratory analyses

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Exploratory: n-back response times for correct targets, averaged within participants before summarizing across participants.

valence	n_back	Mean RT (ms)	SD of participant means (ms)	Participants
negative	1	572	144	5
negative	2	654	121	5
negative	3	691	160	4
neutral	1	532	136	5
neutral	2	612	140	5
neutral	3	632	171	5
positive	1	594	158	5
positive	2	555	111	5
positive	3	518	97	5

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Discussion

Summary of replication attempt

Guidance

Open with one paragraph summarizing the result of the key analysis and your assessment of whether the original finding replicated, partially replicated, or failed to replicate. Base that judgment on the replication assessment above: the size of the effect relative to the original, not just the p value.

Commentary

Guidance

Open-ended commentary, none of it long: (a) what the exploratory analyses suggest; (b) what the replication means, for example whether the differences between your study and the original are definitely, plausibly, or unlikely to be moderators of the result; and (c) any objections or challenges raised by you or the original authors.

Phase 2: Extension

Guidance: choosing your extension

Follow the Extension guidelines page on Canvas. In brief:

- **By default the extension is a best-practices modification of your Phase 1 experiment.** If your replication did not succeed, it is a **rescue**: address the plausible reasons for the failure. If it did succeed, it is an **upgrade**: make the estimate more precise. Typical changes are better or more reliable measurement, more trials, a within-participant design where that is legitimate, comprehension and manipulation checks, clearer instructions, or a larger or better-targeted sample.
- **If your scientific extension pitch was approved at Milestone 3 and your replication succeeded, propose that instead:** the original conditions rerun unchanged, plus the one added condition or manipulation, with a power analysis for the new comparison.
- On either track you may add exploratory measures **at the end** of the study.
- Every change is listed and tagged by category below; everything else stays identical.

State your track on the line below, and delete the others.

Track: Best-practices upgrade / Best-practices rescue / Scientific extension (approved at Milestone 3)

Rationale

Guidance

One or two paragraphs. For a rescue: what are the plausible reasons Phase 1 did not reproduce the original result, and how does each change address one of them? For an upgrade: where is the Phase 1 estimate least precise or least trustworthy, and how will the changes improve it? For a scientific extension: what question does the added condition answer, and why does it follow from the original finding?

Changes from Phase 1

Guidance

Every change from the Phase 1 experiment, one per row, each tagged with a category: Measurement, Trials, Design, Checks, Instructions, Sample, Added condition (scientific track only), or Exploratory measure (end of study only). If it is not in this table, it does not change: same materials, wording, procedure, exclusion rule and analysis. The rows below are examples for the demo; replace them with yours.

#	Category	Phase 1	Phase 2	Why
1	Sample	27 analysed participants	300 analysed participants (150 per frame)	Phase 1's interval was too wide to distinguish the original effect from a much smaller one.
2	Checks	No comprehension check	After the choice, participants say how many people are at risk (600)	Confirms participants read the scenario; failures are reported, and excluded only if preregistered.
3	Exploratory measure	None	At the end: "How confident are you in your choice?" (1–7)	Exploratory only; collected after every confirmatory measure.

Everything not listed above is identical to Phase 1.

Power and sample

Guidance

Justify the Phase 2 sample, and **state how you expect the estimate and its precision to compare with Phase 1**. For an upgrade, plan the sample around the confidence interval you want; for a rescue, around the power to detect the original effect, or the smallest effect you would still consider a successful replication; for a scientific extension, around the power to detect the new comparison. Set `phase2_n_per_group` in the code below; the table uses Phase 1's estimate if you have it and the original's otherwise.

▶ Show code

Analysis plan

Guidance

The confirmatory analysis for Phase 2, written before Phase 2 data collection. On the best-practices tracks this is normally the same key test as Phase 1, run on the Phase 2 data alone; say whether you will also report a pooled estimate across phases, and label it exploratory if so. On the scientific track, specify the test for the new comparison. Name the exclusion rule (unchanged unless it is in the Changes table) and list any exploratory measures.

Run Phase 2 under a new `EXPERIMENT.id` in `experiment.js` (for example `smith2016-phase2`) and set `experiment_id_phase2` at the top of this file to match, so the two phases' data never mix.

Results

▶ Show code

▶ Show code

Discussion

Guidance

Compare Phase 2 with Phase 1 and with the original, using the Key statistics table and the figure above. Did the precision improve as you expected? For a rescue, did addressing the plausible reasons recover the effect? For a scientific extension, what does the new condition show? End with what the two phases together say about the original finding.

Data and code availability

Guidance

Edit the statement below to match what you are actually sharing. Anonymized data and all code should be public; if something cannot be shared, say what and why.

The experiment code, the anonymized data, and the code that produces every number in this report are available at https://github.com/psych251/kopf2013/. The experiment itself can be taken at https://psych251.github.io/kopf2013/. The data contain no names, email addresses, IP addresses, or Prolific IDs; the file linking participants to their Prolific IDs is kept privately for payment and is not shared.